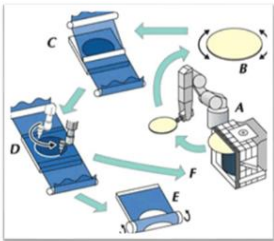


1. Wafer Lamination 공정의 정의
- BG 공정 진행 전 Wafer Top(Pattern)면 BG Tape를 붙이는 공정으로, 외부로부터의 충격이나 오염을 막기위해 필요한 공정이다.

Unit Process	Lamination		
Maker	LINTEC	CUON	DYNATEC
Mdoel Name	RU-3520	CUMA	DT-2030
Image			



Lamination Sequence Flow

2. Material(BG Tape) Information Details

1) LINTEC BG TAPE											
Model	Adhesive Layer	Base Film	Total Thickness	UV Intensity [mW/cm ²]	Exposure Dose [mJ/cm ²]	Peeling Angle[°]	Peeling Speed[mm/s]	Shelf Life	Storage Temperature	Wafer Type별 적용 구분	
E-6142S	20um	110um	130 ± 5um	120 or more (230)	70 or more (190)	155	1.0 / 2.0 / 5.0	12 Month	23±5 ℃ [Room Temp.]	Universal Type	
E-6152S	40um	110um	150 ± 5um	120 or more (190)	70 or more (190)	45	1.0 / 2.5 / 5.0	12 Month	23±5 ℃ [Room Temp.]	Universal Type	
E-8310LS342F	40um	270um	310 ± 9um	150 or more (190)	300 or more (380)	45	1.0 / 2.5 / 5.0	12 Month	23±5 ℃ [Room Temp.]	Universal Type / Ink Dot height under 10um	
E-8180HR	20um	160um	180 ± 5um	150 ~ 250 (230)	300 ~ 500 (380)	45	1.0 / 2.5 / 5.0	7 Month	23±5 ℃ [Room Temp.]	Thin wafer(Normal BG) thickness under 100um	
E-3125BN	20um	105um	125 ± 5um	150 ~ 250 (230)	300 ~ 500 (380)	45	1.0 / 2.0 / 5.0	6 Month	23±5 ℃ [Room Temp.]	Thin/Ultra thin [DBG/SDBG(GAL)용] thickness under 60um	
E-3145KN	40um	105um	145 ± 5um	150 ~ 250 (230)	300 ~ 500 (380)	45	1.0 / 2.0 / 5.0	6 Month	23±5 ℃ [Room Temp.]	Thin/Ultra thin [DBG/SDBG(GAL)용] thickness under 60um	
E-9481G-A	50um	430um	480 ± 5um	150 ~ 250 (230)	570 or More	45	1.0 / 2.0 / 2.0	6 Month	23±5 ℃ [Room Temp.]	Wafer with deep scribe lane or grooves in wafer top pattern	

2) MITSUI BG TAPE											
Model	Adhesive Layer	Intermediate Layer	Base Film	Total Thickness	Exposure Dose [mJ/cm ²]	Peeling Angle[°]	Peeling Speed[mm/s]	Shelf Life	Storage Temperature	Wafer Type별 적용 구분	
HT-440HBA	40um	350um	50um	440 ± 20um	1080 ~ 1200	45	1.0 / 2.0 / 2.0	12 Month	23±5 ℃ [Room Temp.]	Apply for bump wafer (Bump height under 250um)	
HT-230HV	10um	100um	120um	230 ± 8um	1080 ~ 1200	45	1.0 / 2.0 / 2.0	5 Month	23±5 ℃ [Room Temp.]	Apply for bump wafer (Bump height under 90um)	
SB-340HV	20um	200um	120um	340 ± 10um	1080 ~ 1200	45	1.0 / 2.0 / 2.0	5 Month	23±5 ℃ [Room Temp.]	Apply for bump wafer (Bump height under 150um)	
SB-170HM	70um	-	100um	170 ± 7um	-	45	1.0 / 2.0 / 2.0	12 Month	23±5 ℃ [Room Temp.]	Non UV Type / Apply for bump wafer (Bump height under 40um)	
MY-595PT	10um	510um	75um	595 ± 15um	1080 ~ 1200	45	1.0 / 2.0 / 2.0	12 Month	23±5 ℃ [Room Temp.]	Apply for bump wafer (Bump height under 250um)	
HT-230HC	10um	100um	120um	230 ± 8um	1080 ~ 1200	45	1.0 / 2.0 / 2.0	12 Month	23±5 ℃ [Room Temp.]	Apply for bump wafer (Bump height under 90um)	

3) FURUKAWA BG TAPE											
Model	Adhesive Layer	Base Film	Total Thickness	UV Intensity [mW/cm ²]	Exposure Dose [mJ/cm ²]	Peeling Angle[°]	Peeling Speed[mm/s]	Shelf Life	Storage Temperature	Wafer Type별 적용 구분	
SP-594M	30um	100um	130 ± 7um	80 or more	500 ~ 1000	45	1.0 / 2.0 / 5.0	12 Month	23±5 ℃ [Room Temp.]	Universal Type / Wafer thickness above 100um	
SP-5104B	20um	110um	130 ± 7um	80 or more	~ 500	45	1.0 / 2.0 / 5.0	12 Month	23±5 ℃ [Room Temp.]	Universal Type / Wafer thickness above 50um	

3. Standard Recipe for Lamination machine

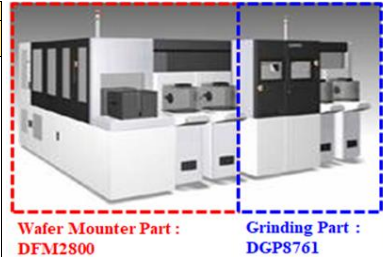
1) Standard range by wafer type			
Wafer Type	Normal Wafer	WLCSP Wafer	COB
Cutting Angle(°)	56~75	-	56~75
Roller Pressure(MPa)	0.4±0.1	0.15~0.45	0.5±0.1
Roller Speed(mm/s)	130±10	3	110±10
Roller Temp.(℃)	20±5	20±5	20±5
Air Pressure(kg/cm ²)	5±1	-	-
Cutter Blade Heater Temp.(℃)	20±10	60~80	120±5
Chuck Table Temp.(℃)	20±10	75±5	20±5
Cutter Speed(mm/s)	1~20	1~20	2±0.5

■ Back grinding process

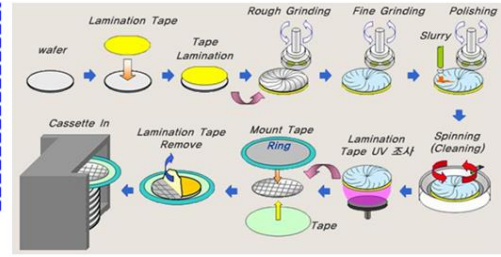
1. Definition of Wafer Back grinding

-. 미세한 크기의 다이아몬드 입자가 담긴 BG Wheel을 이용해 웨이퍼 뒷면을 목표 두께까지 갈고 열전도도를 높이는 공정이다.

Maker	Accretech(TSK)	DISCO
Model	PG300RM	DGP8760+DFM2700 DGP8761+DFM2800
Appreance		
Remark	Polishing Type : Wet polishing (Use the Polishing pad and slurry)	Polishing Type : Dry polishing (Using dry polishing wheel)



Back Grinding Machine



Lamination + Back grinding Full Process

2. Material Information Details

1) Wafer backgrinding wheels and polishing pad information

Item / Steps	Wheel model	Maker	Specification	Expire date	Alarm limit		Remark
					Pre-alarm	Limit	
G1 (Rough grinding)	SDC320- BT300 -50	DISCO	#325 / Resin bond	5years	0.55mm	0.50mm	
	GF13- SD340 -BLC07-50	DISCO	#340 / Resin bond	5years	0.55mm	0.50mm	
	RSL (BG01-T2-300D:RSL)	EHWA	#325 / Resin bond	5years	0.55mm	0.50mm	
	VCS	Shin Han	#800 / Vitrified bond	5years	1.0mm	0.80mm	
G2 (Fine grinding)	SD4800- VV6B -50	DISCO	#4800 / Vitrified bond	5years	0.55mm	0.50mm	
	VV3A	DISCO	#6000 / Vitrified bond	5years	0.55mm	0.50mm	
	VK8B	EHWA	#5000 / Vitrified bond	5years	0.55mm	0.50mm	
	TF9	Shin Han	#8000 / Vitrified bond	5years	1.0mm	0.80mm	
	VP1	Shin Han	#12000 / Vitrified bond	5years	1.7mm	1.50mm	
Wet Polishing pad (TSK M/C)	TP300V	Accretech	Thickness : 1.2 ~ 1.34mm	6month	180wafer	200wafer	
	IP300V4	INT solution	Thickness : 1.2 ~ 1.34mm	6month	180wafer	200wafer	
Dry Polish Wheel (Disco M/C)	450-11T-60-DPW-026:DP-F05	DISCO	Dry polishing wheel	3years	1.10mm	1.00mm	
	DPEG-GA0001(GDP)	DISCO	Gettering DP wheel	3years	1.10mm	1.00mm	
	DP08	DISCO	Dry polishing wheel	3years	1.10mm	1.00mm	
	TS200L	Accretech	Abrasive slurry	6month	Low limit alarm		
Slurry	SI500C5	제일 모직	Ultra High purity	1years	Low limit alarm		

2) Dicing Tape Information

* Die Attach 공정에 Epoxy 접착제 사용이 요구되는 경우

Maker	Model	Adhesive Layer	Base Film	Total Thickness	UV Intensity [mW/cm ²]	Exposure Dose [mJ/cm ²]	Mount Temperature		Storage Temperature	Remark
							Maker Recommend	Actual Use		
Lintec	D-175	10um	80um	90 ± 10um	120 or more (230)	70 or more (190)	50 ± 5℃ (50℃)	50 ± 5℃ (50℃)	23±5℃ [Room Temp.]	Universal Type (PVC Dicing tape)
Lintec	D-174	10um	80um	90 ± 10um	-	70 or more	50 ± 5℃ (50℃)	50 ± 5℃ (50℃)	23±5℃ [Room Temp.]	Universal Type D-175 대비 접착력이 높고, Bank Mark가 없음 - Back Side Chipping에 유리함
Lintec	D-485H	5um	80um	85 ± 10um	230	190	Room temp.	23 ± 5℃	23±5℃ [Room Temp.]	Universal Type (PO Dicing tape) D-175 대비 접착력이 높고, Bank Mark가 없음 - Back Side Chipping에 유리함
Lintec	D-675	10um	80um	90 ± 10um	120 or more (230)	70 or more (190)	Room temp.	23 ± 5℃	23±5℃ [Room Temp.]	Universal Type (PO Dicing tape)

3) DAF(Die Attach Film) Information

* Die Attach 공정에 Attach Film의 적용이 요구되는 경우

Maker	Model	Adhesive Layer	Sticky Layer	Base Film	Total Thickness	UV Intensity [mW/cm ²]	Exposure Dose [mJ/cm ²]	Tape Mount Temperature		Shelf Life (Below 5℃)	Shelf Life (After aging on 25℃)	Aging Time	Remark
								Maker Recommend	Actual Use				
Hitach	FH-M526S(T)	50um	20um	90um	160um	10 ~ 60	100 ~ 200	65 ± 5℃(65℃)	75 ± 5℃(75℃)	12 Month	28 days	Over 4hr@R.T.	FOW Type
Hitach	FH-7221S(T)	20um	10um	90um	120um	30 ~ 90	100 ~ 200	65 ± 5℃(65℃)	75 ± 5℃(75℃)	12 Month	28 days	Over 4hr@R.T.	Normal type
Hitach	FH-9049S(T)	40um	10um	150um	200um	30 ~ 90	100 ~ 200	65 ± 5℃(65℃)	70 ± 5℃(70℃)	12 Month	28 days	Over 4hr@R.T.	Normal type
Hitach	FH-7134S(T)	10um	15um	90um	115um	30 ~ 90	100 ~ 200	65 ± 5℃(65℃)	75 ± 5℃(75℃)	12 Month	28 days	Over 4hr@R.T.	Normal type
Hitach	FH-7145S(T)	10um	10um	90um	110um	30 ~ 90	100 ~ 200	65 ± 5℃(65℃)	75 ± 5℃(75℃)	12 Month	28 days	Over 4hr@R.T.	Normal type
Hitach	FH-R526S(T)	50um	20um	90um	160um	10 ~ 60	100 ~ 200	65 ± 5℃(65℃)	75 ± 5℃(75℃)	12 Month	28 days	Over 4hr@R.T.	FOW Type
Hitach	HR-9104ST	10um	30um	80um	120um	Skip (Non-UV)		65 ± 5℃(65℃)	75 ± 5℃(75℃)	12 Month	28 days	Over 4hr@R.T.	Normal type / Non-UV
Hitach	HR-C204ST	20um	30um	80um	130um	Skip (Non-UV)		65 ± 5℃(65℃)	75 ± 5℃(75℃)	12 Month	28 days	Over 4hr@R.T.	For small Die / Non-UV
LG CHEMICAL	LDA-5A23T-10	10um	10um	100um	120um	50 ~ 70	50 ~ 300	75 ± 5℃(75℃)	70 ± 5℃(70℃)	12 Month	1 Month	Over 4hr@R.T.	
LG CHEMICAL	LDA-7A24T-60	60um	10um	110um	180um	-	300	75 ± 5℃(75℃)	70 ± 5℃(70℃)	6 Month	3 weeks	Over 4hr@R.T.	
LG CHEMICAL	LD-RC56ST	110um	10um	100um	220um	-	200~400	65 ± 5℃(65℃)	65 ± 5℃(65℃)	6 Month	4 weeks	Over 4hr@R.T.	FOD Type
LG CHEMICAL	LDA-5A23T-20	20um	10um	100um	130um	-	300	70 ± 5℃(70℃)	70 ± 5℃(70℃)	12 Month	4 weeks	Over 4hr@R.T.	
LG CHEMICAL	LDA7A24T-50	50um	10um	100um	160um	-	300	75 ± 5℃(75℃)	70 ± 5℃(70℃)	6 Month	3 weeks	Over 4hr@R.T.	
LG CHEMICAL	LD-N23E(T)-20	20um	10um	100um	130um	-	300	65 ± 5℃(65℃)	65 ± 5℃(65℃)	12 Month	1 Month	Over 4hr@R.T.	
LG CHEMICAL	LD-A268ST	20um	10um	90um	120um	-	200 ~ 400	65 ± 5℃(65℃)	65 ± 5℃(65℃)	12 Month	1 Month	Over 4hr@R.T.	
INNOX	WL0020-02DS	20um	10um	100um	130um	60 ~ 80	200	70 ± 10℃(70℃)	75 ± 5℃(75℃)	12 Month	2 Month	Over 4hr@R.T.	For Normal Die Wafer (only Blade 가능)
INNOX	IDU0C1-20E(T)	20um	10um	100um	130um	60 ~ 80	200	65 ~ 70℃	65 ± 5℃(65℃)	6 Month	28 days	Over 4hr@R.T.	For Small Die Wafer
INNOX	IDU0B3L-20T(E)	20um	10um	100um	130um	80	200	65 ~ 70℃	65 ± 5℃(65℃)	6 Month	28 days	Over 4hr@R.T.	For Normal Die Wafer (Blade & NDBG & SDBG 가능)
INNOX	ICA0A2-43	DAF 30um(A:10+B:20) / PI:13um	10um	100um	153um	-	200	75 ± 5℃(75℃)	75 ± 5℃(75℃)	12 Month	1 Month	Over 4hr@R.T.	CAL(Compensation Adhesive Layer)

3. Standard guidance to make a BG Machine recipe

* 각 파라미터에 기준은 괄호 안의 값으로 정하며, 필요한 경우 표시된 범위 내에서 변경된 값을 적용할 수 있다.

* 단, 특별한 문제가 없는 경우에는 대표 표준값을 적용한다.

* 특별한 경우 : Current over (Z1 / Z2), Wafer Crack, Wafer burning, Corner crack, Wafer edge crack 등

* Bump Wafer의 Off-set Height 설정은 Lamination Sheet 참조

1) Back Grinding Condition for each wafer type & Sawing Method

① 일반 Wafer + Stealth Dicing 적용

Customer	Wafer Size	Wafer Type	Sawing Type	NCG	Grinding Step	Coolant	BG Wheel Type	Wheel RPM	Cut Step	Feed Rate(um/s)	Grinding Amount	Chuck RPM	
												8761	8760
Check	12Inch	일반 Wafer	Stealth Dicing	Option	G1	Wheel + Inner	VC5	3200	P1	5±1(5)	-	300	-300
									P2	3±1(3)	50	300	-300
									P3	2±1(2)	50	300	-300
					G2	Inner Only	VP1	3000	P1	0.4±0.1(0.4)	45	-300	-300
									P2	0.3±0.1(0.3)	10	-300	-300
									P3	0.2~0.3(0.2)	5	-300	-300

② Bump Wafer + Stealth Dicing 적용

Check	12Inch	Memory (Bump Wafer/ FCBOC)	Stealth Dicing	Option	G1	Wheel + Inner	VC5	2700	P1	5±1(5)	-	300	-300
									P2	3±1(3)	50	300	-300
									P3	2±1(2)	50	300	-300
					G2	Wheel + Inner	VP1	2700	P1	0.4±0.1(0.4)	10	-300	-300
									P2	0.3±0.1(0.3)	10	-300	-300
									P3	0.2~0.3(0.2)	10	-300	-300

③ 일반 Wafer + Blade Dicing 적용

Check	12Inch	일반 Wafer	Blade Dicing	Option	G1	Wheel + Inner	VC5	3200	P1	4±1(4)	-	300	-300
									P2	3±1(3)	50	300	-300
									P3	2±1(2)	50	300	-300
					G2	Wheel + Inner	VP1	3000	P1	0.4±0.1(0.4)	25	-300	-300
									P2	0.3±0.1(0.3)	10	-300	-300
									P3	0.2~0.3(0.2)	5	-300	-300

④ LSI Wafer + Blade Dicing 적용

Check	12Inch	LSI (No bumped)	Blade Dicing	Option	G1	Wheel + Inner	BT300	3200	P1	5±1(5)	-	300	-300
									P2	3±1(3)	100	300	-300
									P3	2±1(2)	50	300	-300
					G2	Inner Only	VV3A	3000	P1	0.4±0.1(0.4)	10	-300	-300
									P2	0.3±0.1(0.3)	10	-300	-300
									P3	0.2~0.3(0.2)	10	-300	-300

⑤ Bump Wafer + Blade Dicing 적용

Check	8Inch 12Inch	Non-Memory (Bump Wafer)	Blade Dicing	Not Use	G1	Wheel + Inner	BT300 / RSL	2500	P1	5±1(5)	-	300	-300
									P2	3±1(3)	50	300	-300
									P3	2±1(2)	30	300	-300
					G2	Inner Only	VV3A / VK8B	2700	P1	0.5±0.1(0.5)	10	-300	-300
									P2	0.3±0.1(0.3)	10	-300	-300
									P3	0.2±0.1(0.2)	10	-300	-300

2) Dry Polishing Condition

* Polishing 은 Wafer Taget Thickness 가 ≤60um 이거나, 고객의 요청이 있는 경우 진행한다.
* 그 외 특이사항이 없는 경우 Polishing은 생략해도 좋다.

Wafer Size	12Inch	Spindle	1500	Air Cut Amount / Cut Feed Rate		100 / 10
Sequence	Seq.1	Seq.2	Seq.3	Seq.4	Seq.5	
Feed Rate Down	0.5	0.3	0.3	0	0.3	
Feed Rate Up	1.0	1.0	1.0	1.0	1.0	
Chuck Table RPM	300	300	300	300	300	
Process Position	140	140	135	140	140	
Process Load	100	150	150	150	150	
Wating Time	10	0	40	0	0	
X Feed Rate	-	0.1	0.1	0.1	0	
Swing Time	-	1		1	1	
Polishing Time	150 Sec					

3) Tape Mount Parameter

Tape Mount	Inspection Table Align	모든 Wafer에 적용
	Attachment table Temp. (℃)	Mount Tape 정보 참조
	Attachment Speed (mm/s)	20~30
	UV irradiation Speed (mm/s)	15~20
BG Tape Remove (Peeling)	Peeling Bar Temp. (℃)	아래 Peeling Bar 온도 기준 참조
	Peeling Table Temp. (℃)	23±5℃ (Room Temp.)
	BG Tape Peeling Speed (mm/s)	BG Tape (Lamination Sheet) 정보 참조
	BG Tape Peeling Angle (°)	BG Tape (Lamination Sheet) 정보 참조

4) Peeling bar 온도 기준

Classify by BG tape	Temperature(℃)	Remark
EL-100R removal tape 적용된 경우 (All BG Tape)	235 ± 5	-
EL-100R 미 적용_E6142, E6152, E8180HR, E8310 (Lintec)	180 ± 5	-
E3125BN, E3145KN (Lintec)	For DBG Wafer : 165 ± 5 For GAL Wafer : 165 ± 15	-
SP-594M, SP-5104B (Furukawa)	180 ± 5	-
HT440, HT230, SB340, SB170, MY595 (Mitsui)	180 ± 5	-

Wafer Size	8Inch	Spindle	1800	Air Cut Amount / Cut Feed Rate		Min 50 / 10
Sequence	Seq.1	Seq.2	Seq.3	Seq.4	Seq.5	
Feed Rate Down	1.0	1.0	1.0	1.0	1.0	
Feed Rate Up	1.5	1.5	1.5	1.5	1.5	
Chuck Table RPM	300	300	300	300	300	
Process Position	100	100	100.5	150	150	
Process Load	100	150	150	150	150	
Waiting Time	5	0	0	0	0	
X Feed Rate	-	0.1	0.1	0	0	
Swing Time	-	1	12	1	1	
Polishing Time	120 Sec					

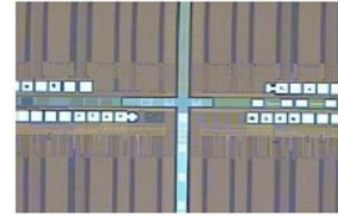
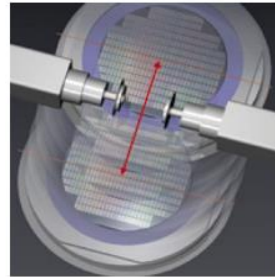
■ Wafer Sawing (Wafer Dicing)

* Wafer Sawing 참조 동영상 : <https://youtu.be/SWtQD8wBtYQ>

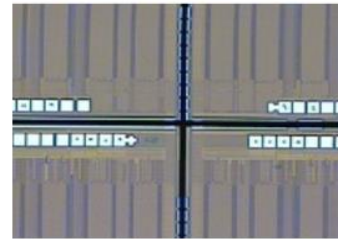
1. Definition of Wafer Sawing

- Wafer sawing은 Die Attach 전 Diamond Blade를 이용하여 Wafer를 자르고 개별 Die로 분리하는 공정이다.

Maker	DISCO
Model	DFD6360 / DFD6361 / DFD6362
Appreanance	
Remark	The higher the version, the more functions are added.



Before Sawing



After Sawing

2. Saw Recipe 표준 가이드

※ Recipe Setup 시 고객의 요청을 우선하며, 별도의 요청이 없는 경우 자사 표준을 따른다.

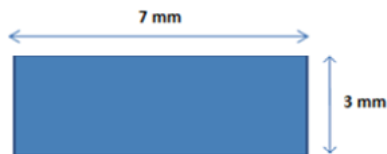
1) Sawing 방식의 선택 기준

- Wafer 특성 또는 고객의 요청에 따라 방식은 바뀔 수 있음
- Wafer 선평(Si node)이 65nm 이하의 Low-K Wafer인 경우 Laser Grooving 공정의 적용을 반드시 검토해야 한다.

Scribe Line Width	Sawing Method	Wafer Thickness	Cut Method
Min. 60um	Blade Sawing	50~100um	Full Cut
		100um ↑	Step Cut
Min. 60um	Stealth Dicing	Min. 40um	Grind After Laser Saw

2) Wafer Cutting 순서의 결정 기준

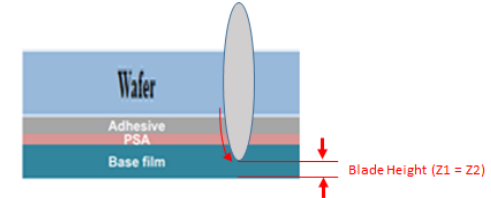
- 기본적으로는 Die의 긴변을 먼저 자르도록 셋팅한다.
- 만약, Wafer X축, Y축의 Scribe Lane 폭이 서로 다른 경우 폭이 좁은 축을 먼저 Cutting 해야 Chipping 예방에 유리하다.
- 만약 Die의 길이 비율이 0.5 이하인 Wafer를 Sawing 할 경우, 짧은 변을 먼저 Cutting 하는 것이 Chipping 예방에 유리하다.
- 만약, Scribe Lane with가 넓어서 두 줄을 Cutting해야 하는 경우는 두 줄 Cutting 을 우선하는 것이 Chipping, Flying Die에 유리하다.



※ Die Ratio = 짧은 변 길이 / 긴변 길이
Ex) $0.428571 = 3\text{mm} / 7\text{mm}$

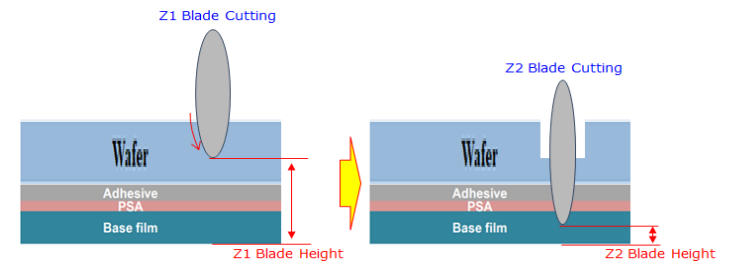
1) Full Cut Saw

Z1과 Z2 Blade Model이 서로 같으며 한번에 원하는 깊이까지 Cutting



2) Step Cut Saw

Z1과 Z2 Blade Model이 서로 다르며, Z1으로 일부 Cutting 후 Z2 Cutting



3) Blade 선정 기준

a) 기본 조건

- 고객 요청이 있을 경우 요청에 우선한다. (단, 요청이 불합리 또는 Risk가 있다고 판단되는 경우 협의 진행한다.)
- Blade 선정은 Wafer 특성, 고객 지침, 테스트 결과에 따라 변경될 수 있다.

Wafer Thickness	Scribe line width	Cut Method	Z1 Grit	Z1 Blade	Z1 Concentration	Z2 Grit	Z2 Blade	Z2 Concentration	Remark
60um ↓	60um ≤	Full Cut	#(Mesh) 4000 ~	B or C Kerf	50 ~	#(Mesh) 4000 ~	B or C Kerf	50 ~	Use same blade both Z1 and Z2
	60um >	Step Cut	#(Mesh) 3500 ~	D Kerf	50 ~	#(Mesh) 3000 ~	B or C Kerf	50 ~	
60 ~ 100um ↓	60um ≤	Full Cut	#(Mesh) 3000 ~	A ~ C Kerf	50 ~	#(Mesh) 3000 ~	A ~ C Kerf	50 ~	Use same blade both Z1 and Z2
	60um >	Step Cut	#(Mesh) 3500 ~	D or E Kerf	50 ~	#(Mesh) 3000 ~	B or C Kerf	50 ~	
100 ~ 250um ↓	60um ≤	Full or Step Cut	#(Mesh) 3000 ~	B ~ D Kerf	50 ~	#(Mesh) 2000 ~	A or B Kerf	50 ~	Cu method 평가 필요
	60um >	Step Cut	#(Mesh) 3000 ~	D or E Kerf	50 ~	#(Mesh) 2000 ~	B or C Kerf	50 ~	
250 ~ 500um ↓	60um ≤	Full or Step Cut	#(Mesh) 3000 ~	C ~ E Kerf	50 ~	#(Mesh) 2000 ~	B ~ D Kerf	50 ~	Full Cut 이 필요한 경우 Laser Grooving 추가 검토 필요
	60um >	Step Cut	#(Mesh) 3000 ~	D ~ F Kerf	50 ~	#(Mesh) 2000 ~	B ~ E Kerf	50 ~	
500um ~	60um ≤	Step Cut	-	-	-	-	-	-	
	60um >	Step Cut	#(Mesh) 3000 ~	D ~ F Kerf	50 ~	#(Mesh) 2000 ~	B ~ E Kerf	50 ~	

b) Blade 폭(Kerf Width) 선정 기준

㉑ 기본 조건 : Step Cut을 적용하는 경우 Z1 Blade와 Z2 Blade 의 kerf Width는 2단계 이상 차이로 선정한다.

㉒ Wafer 특성 및 ENG'R 검토 결과에 따라 Blade 선정 기준은 달라질 수 있으나 기본적인 기준을 우선으로 한다.

㉓ Sawing Lane 폭 > 60um 인 경우 (Wide Sawing Lane)

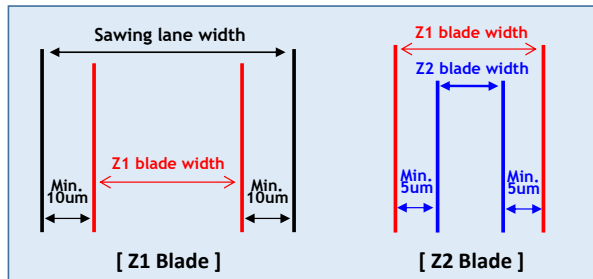
- Sawing Lane 폭 - Z1 Blade kerf 폭 ≥ Sawing Lane Margin 최소 10um
- Z1 Blade kerf 폭 - Z2 Blade kerf 폭 ≥ Sawing Lane Margin 최소 5um

㉔ Sawing Lane 폭 ≤ 60um 인 경우 (Narrow Sawing lane)

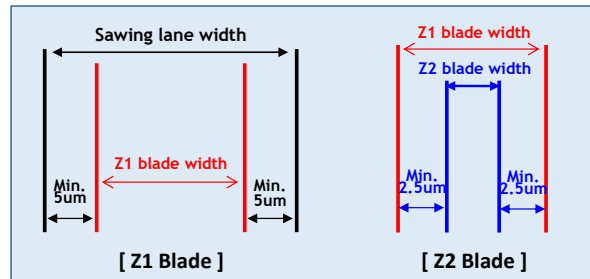
- Sawing lane 폭 - Z1 Blade kerf 폭 ≥ Sawing Lane Margin 최소 5um
- Z1 Blade kerf 폭 - Z2 Blade kerf 폭 ≥ Sawing Lane Margin 최소 2.5um

㉕ Special case : For NXP社

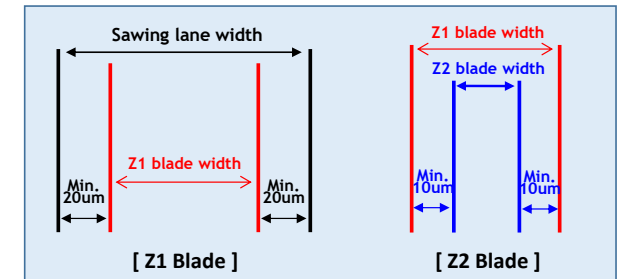
- Sawing lane 폭 - Z1 Blade kerf 폭 ≥ Sawing Lane Margin 최소 20um
- Z1 Blade kerf 폭 - Z2 Blade kerf 폭 ≥ Sawing Lane Margin 최소 10um



[㉓ Wide sawing lane]



[㉔ Narrow sawing lane]



[㉕ For NXP]

c) Blade Size 선정 기준 (사용 가능한 Exposure+Kerf Width 조합표)

- Target Wafer의 Saw Lane Width와 Thickness Data를 확인 후 검토한다.
- Blade Exposure는 Sawing이 진행될 Wafer Thickness 를 검토하여 가능한 오래 사용할 수 있는 조건으로 검토한다.
- 적용 가능한 Blade 종류는 Sawing Blade List 참조

Kerf width	Exposure mm	A 0.38~0.51	B 0.51~0.64	C 0.64~0.76	D 0.76~0.89	E 0.89~1.02	F 1.02~1.15
A	0.015~0.02	AA	X	X	X	X	X
B	0.02~0.025	AB	BB	CB	X	X	X
C	0.025~0.03	X	BC	CC	DC	X	X
D	0.03~0.035	X	BD	CD	DD	ED	FD
E	0.035~0.04	X	X	CE	DE	EE	FE
F	0.04~0.05	X	X	CF	DF	EF	FF
G	0.05~0.06	X	X	X	X	EG	FG

4) Blade Height 선정 기준

a) Z1 Blade Height

- Z1 Blade Height 는 Chuck Table Top을 영점 기준으로 하고 Z1 Blade Cut 하는 깊이까지를 말한다.
- Z1 Blade Height = Wafer Thickness + Tape Thickness - Z1 Cut Depth

Wafer Thickness	Scribe line width	Cut Method	Z1 Cut depth
60um ↓	60um ≤	Full Cut	Die 두께의 30~50%
	> 60um	Step Cut	Die 두께의 30~50%
60 ~ 100um ↓	60um ≤	Full Cut	Die 두께의 30~50%
	> 60um	Step Cut	Die 두께의 30~50%
100 ~ 250um ↓	60um ≤	Full or Step Cut	Max.120um 또는 그 이하
	> 60um	Step Cut	Max.120um 또는 그 이하
250 ~ 500um ↓	60um ≤	Full or Step Cut	Max.150um 또는 그 이하
	> 60um	Step Cut	Max.150um 또는 그 이하
500um ~	60um ≤	Step Cut	ENG'R 평가 필요
	> 60um	Step Cut	Die 두께의 30% 이하

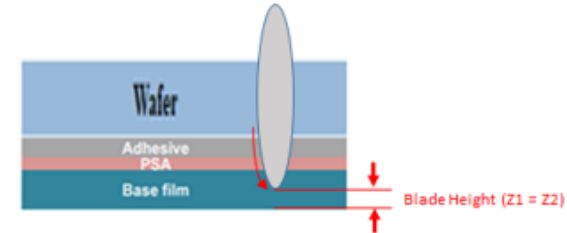
b) Z2 Blade Height

- 표준 Z2 Blade Cut Dpth : Tape 총 두께(Base film + Adhesive layer)의에서 25±10um
- Z2 blade cut height는 Mount Tape의 종류 및 아래 표를 참조하여 설정한다.

Tape model	Blade height(um)	Tape model	Blade height(um)
D-175	70±10	FH-RC13ST	110±10
D-485H	70±10	ICA0A2-43	80±10
FH, HR Series	70~105	LDA-7A24T Series	80±10
SG200 Series	65±10	LDA-5202 Series	80±10
HR-9XXX Series	95±5	FH-9409ST	145±20
WL0020 Series	80±10	IDU0B3L-20T	80±10
IDU0C1-20	80±10	LD-N23E-20	80±10
HR-D206ST	90±10	LDA-5A23T Series	80±10
HR-C204ST	90±10	LD-RC56ST	75±10
ICA0B2-43	80±10		

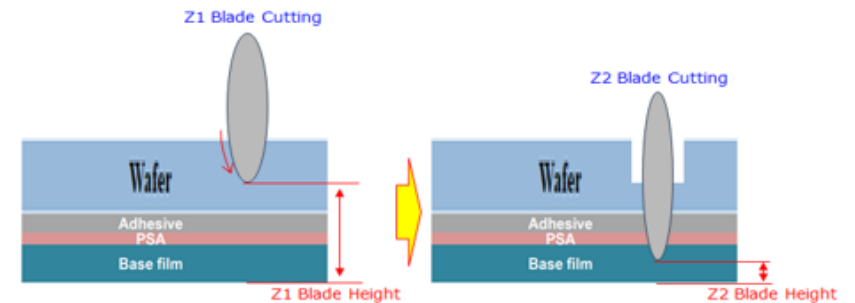
1) Full Cut Saw

Z1과 Z2 Blade Model이 서로 같으며 한번에 원하는 깊이까지 Cutting



2) Step Cut Saw

Z1과 Z2 Blade Model이 서로 다르며, Z1으로 일부 Cutting 후 Z2 Cutting



■ Wafer Sawing (Wafer Dicing)

1. Blade List

Vendor	Model	O.D (Outer diameter)	LD (Inner diameter)	Exposure	Thickness
Disco	NBC-ZH-204J 27HDCD	55.56 ± 0.05	19.055 ± 0.005	640~760um	30~35um
Disco	NBC-ZH-127F 27HCBC	55.56 ± 0.05	19.055 ± 0.005	510~640um	25~30um
Disco	NBC-ZH-127J 27HCBB	55.56 ± 0.05	RU-3520	510~640um	20~25um
Disco	ZH05-SD3000-N1-90CC	55.56 ± 0.05	19.055 ± 0.005	640~760um	25~30um
Disco	NBC-ZH-203J 27HCCE	55.56 ± 0.05	19.055 ± 0.005	640~710um	35~40um
Disco	ZH05-SD2000-N1-70ED	55.56 ± 0.05	19.055 ± 0.005	890~1,020um	30~35um
Disco	ZH05-SD4000-N1-70DD	55.56 ± 0.05	19.055 ± 0.005	760~890um	30~35um
Disco	ZH05-SD3500-N1-50BB	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~25um
Disco	ZHZZ-SD4800-H1-50AA	55.56 ± 0.05	19.055 ± 0.005	380~510um	15~20um
Disco	ZH05-SD3500-N1-50CC	55.56 ± 0.05	19.055 ± 0.005	640~760um	25~30um
Disco	ZHZZ-SD3000-H1-50BA	55.56 ± 0.05	19.055 ± 0.005	510~640um	15~20um
Disco	ZH05-SD3000-N1-90CE	55.56 ± 0.05	19.055 ± 0.005	640~760um	35~40um
Disco	ZH05-SD3000-N1-50CC	55.56 ± 0.05	19.055 ± 0.005	640~760um	25~30um
Disco	ZH05-SD4000-N1-50CC	55.56 ± 0.05	19.055 ± 0.005	640~760um	25~30um
Disco	ZH05-SD3500-N1-50AA	55.56 ± 0.05	19.055 ± 0.005	380~510um	15~20um
Disco	ZH05-SD4000-N1-50DD	55.56 ± 0.05	19.055 ± 0.005	760~890um	30~35um
Disco	ZHT-105F-SE 27HEDD	55.56 ± 0.05	19.055 ± 0.005	760~890um	30~35um
Disco	ZHZZ-SD4800-H1-50BC	55.56 ± 0.05	19.055 ± 0.005	510~640um	25~30um
Disco	ZH05-SD4000-N1-50DD	55.56 ± 0.05	19.055 ± 0.005	760~890um	30~35um
Disco	ZH05-SD3000-N1-70BB	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~25um
Disco	ZH05-SD4000-N1-50 CD	55.56 ± 0.05	19.055 ± 0.005	640~760um	30~35um
Disco	ZH05-SD4800-N1-50 BB	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~25um
Disco	ZH05-SD3500-N1-50 DD	55.56 ± 0.05	19.055 ± 0.005	760~890um	30~35um
Disco	ZH05-SD3500-N1-50 BA	55.56 ± 0.05	19.055 ± 0.005	510~640um	15~20um
Disco	ZH05-SD2000-N1-70 DC	55.56 ± 0.05	19.055 ± 0.005	760~890um	25~30um
Disco	ZH05-SD3000-N1-50 BB	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~25um
Disco	ZH05-SD3500-N1-50 EE	55.56 ± 0.05	19.055 ± 0.005	890~1,020um	35~40um
Disco	ZHZZ-SD4800-H1-50 CC	55.56 ± 0.05	19.055 ± 0.005	640~760um	25~30um
Disco	ZH05-SD2000-N1-70 FE	55.56 ± 0.05	19.055 ± 0.005	1,020~1,150um	35~40um
Disco	ZH05-SD2000-N1-50 FD	55.56 ± 0.05	19.055 ± 0.005	1,020~1,150um	30~35um
Disco	ZH05-SD2000-N1-90 KK	55.56 ± 0.05	19.055 ± 0.005	1,670~1,800um	90~100um
Disco	ZH05-SD3000-N1-50 DF01	55.56 ± 0.05	19.055 ± 0.005	0.760~0.890um	40~45um
Disco	ZH05-SD3500-N1-50 DF02	55.56 ± 0.05	19.055 ± 0.005	760~890um	45~50um
Disco	ZH05-SD3500-N1-50-N3402 DF01	55.56 ± 0.05	19.055 ± 0.005	760~890um	40~45um
Disco	ZH05-SD3500-N1-50 CF02	55.56 ± 0.05	19.055 ± 0.005	640~760um	45~50um
Disco	ZH05-SD3000-N1-50 DD	55.56 ± 0.05	19.055 ± 0.005	760~890um	30~35um
Disco	ZHZZ-SD3000-H1-70 BB01	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~22um
Disco	ZHZZ-SD4800-H1-50 AA01	55.56 ± 0.05	19.055 ± 0.005	380~510um	15~17um
Disco	ZH05-SD3500-N1-50 CF	55.56 ± 0.05	19.055 ± 0.005	640~760um	40~50um
Disco	ZHZZ-SD4500-H1-50-BB-C444	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~25um
Disco	ZH05-SD4500-H1-50-BB-C444	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~25um
Disco	ZH14-SD2000-V1-70-A3100EC	55.56 ± 0.05	19.055 ± 0.005	890~1020um	25~30um
Disco	ZH05-SD3500-N1-50-A3819-CC	55.56 ± 0.05	19.055 ± 0.005	640~760um	28~30um
Disco	ZH05-SD3500-N1-50-A3821-BB	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~22um
Disco	ZH05-SD3500-N1-50-EF 01	55.56 ± 0.05	19.055 ± 0.005	890~1020um	40~45um
Disco	ZH05-SD3500-N1-50-DC	55.56 ± 0.05	19.055 ± 0.005	760~890um	25~30um
Disco	ZH05-SD3000-N1-50-DC	55.56 ± 0.05	19.055 ± 0.005	760~890um	25~30um
Disco	ZH05-SD3500-N1-50-DC	55.56 ± 0.05	19.055 ± 0.005	640~760um	35~38um
Disco	ZHZZ-SD4500-H1-50-BB-C444	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~25um
Disco	ZH05-SD4500-N1-50-BB-C444	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~25um
Disco	ZH05-SD3500-N1-50-A4136 EF-C444	55.56 ± 0.05	19.055 ± 0.005	890~1020um	40~45um
Disco	ZH05-SD3000-N1-50-A3896 DC-C444	55.56 ± 0.05	19.055 ± 0.005	760~890um	25~30um
Disco	ZH05-SD3500-N1-50-A3984 DE-C444	55.56 ± 0.05	19.055 ± 0.005	760~890um	35~40um
Disco	ZH05-SD3000-N1-50-DC-C444	55.56 ± 0.05	19.055 ± 0.005	760~890um	25~30um
Disco	ZH05-SD4800-N1-50-A3818 EF-C444	55.56 ± 0.05	19.055 ± 0.005	890~1020um	40~45um
Disco	ZH05-SD4000-N1-50-A3818 EF-C444	55.56 ± 0.05	19.055 ± 0.005	890~1020um	40~45um
Disco	ZH05-SD3000-N1-50-A3818 EF-C444	55.56 ± 0.05	19.055 ± 0.005	890~1020um	40~45um
Disco	ZHZZ-SD4500-H1-50-A3856 BB-C444	55.56 ± 0.05	19.055 ± 0.005	510~640um	20~25um
Disco	ZH05-SD3500-N1-50-DC-C444	55.56 ± 0.05	19.055 ± 0.005	760~890um	25~30um
Disco	ZH05-SD3500-N1-90-A3907 CE-C444	55.56 ± 0.05	19.055 ± 0.005	640~760um	35~38um
Disco	ZH05-SD4000-N1-50-A3907 CE-C444	55.56 ± 0.05	19.055 ± 0.005	640~760um	35~38um
Disco	ZH05-SD4500-N1-50-A3907 CE-C444	55.56 ± 0.05	19.055 ± 0.005	640~760um	35~38um

Specifications

Grit type	Bond type	Special specification	
SD	N1		
ZH05 - SD 2000 - N1 - 110 - A**** D D			
Grit size	Concentration	Exposure	Kerf width ^{*1}
1500 #1500	50	A 0.38~0.51	A 0.015~0.020
1700 #1700	70	B 0.51~0.64	B 0.020~0.025
1800 #1800	90	C 0.64~0.76	C 0.025~0.030
2000 #2000	110	D 0.76~0.89	D 0.030~0.035
2500 #2500	130	E 0.89~1.02	E 0.035~0.040
3000 #3000		F 1.02~1.15	F 0.040~0.050
3500 #3500		G 1.15~1.28	G 0.050~0.060
4000 #4000		H 1.28~1.41	H 0.060~0.070
4500 #4500		I 1.41~1.54	I 0.070~0.080
4800 #4800		J 1.54~1.67	J 0.080~0.090
5000 #5000		K 1.67~1.80	K 0.090~0.100

(mm) (mm)

^{*1} Can be manufactured to produce a maximum kerf width of 0.2 mm

Standard sizes^{*2}

Exposure		A	B	C	D	E	F	G
Kerf width	mm	0.38-0.51	0.51-0.64	0.64-0.76	0.76-0.89	0.89-1.02	1.02-1.15	1.15-1.28
A	0.015 -0.020	AA	BA					
B	0.020 -0.025	AB	BB	CB				
C	0.025 -0.030		BC	CC	DC			
D	0.030 -0.035		BD	CD	DD	ED		
E	0.035 -0.040			CE	DE	EE	FE	
F	0.040 -0.050			CF	DF	EF	FF	GF
G	0.050 -0.060			CG	DG	EG	FG	GG

^{*2} Please contact a DISCO representative for details.

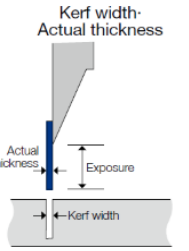
Electroformed Bond Blades with HUB

ZH05 SERIES



Standard concentrations

Grit size	Concentration
	50 70 90 110 130
#1500	✓ ✓ ✓ ✓ ✓
#1700	✓ ✓ ✓ ✓ ✓
#1800	✓ ✓ ✓ ✓ ✓
#2000	✓ ✓ ✓ ✓ ✓
#2500	✓ ✓ ✓ ✓ ✓
#3000	✓ ✓ ✓ ✓ ✓
#3500	✓ ✓ ✓ ✓ ✓
#4000	✓ ✓ ✓ ✓ ✓
#4500	✓ ✓ ✓ ✓
#4800	✓ ✓ ✓ ✓
#5000	✓ ✓ ✓ ✓



Electroformed Bond Hub Blades

NBC-ZH SERIES

Specifications

■Standard Type

Bond ^{*2}	Hub type	Diameter
O Standard Bond	SE Sharp hub edge type	27H O.D. ø55.56
J Soft Bond		LD ø19.05
F For Backside Chipping Reduction		35H O.D. ø78.20
		LD ø31.75 (mm)

[#]NBC - ZH 2 05 0 - SE 27H E D D

Concentration	Grit size	Grit size code	Exposure	kerf width
1 Low Concentration	07 #1500	G	A 0.38~0.51	A 0.015~0.020
	06 #1700	F	B 0.51~0.64	B 0.020~0.025
	29 #1800	C	C 0.64~0.76	C 0.025~0.030
2 Standard Concentration	05 #2000	E	D 0.76~0.89	D 0.030~0.035
	40 #3000	D	E 0.89~1.02	E 0.035~0.040
	03 #3500	F	F 1.02~1.15	F 0.040~0.050
	27 #4000	C	G 1.15~1.28	G 0.050~0.060
	02 #4500	B	H 1.28~1.41	H 0.060~0.070
	26 #4800	A	I 1.41~1.54	I 0.070~0.080
	22 #5000		J 1.54~1.67	J 0.080~0.090
			K 1.67~1.80	K 0.090~0.100
			(mm)	L 0.100~0.110 (mm)

^{*1} Products that include a special specification may be denoted with "ZH1****"

^{*2} All slit widths are 0.5mm (except for the SS type) All slit depths are 80~100% of exposure (except for the SS type) Blade thicknesses greater than 0.06mm are available.

^{*3} Blade thicknesses greater than 0.1mm are available.

■Actual blade thickness

NBC - ZH 2 05 0 - D - S1 - T1 - SE

Exposure	Grit ^{*2}	Blade shape ^{*3}
A 0.38~0.51	S1 No. of slits 4	T1
B 0.51~0.64	S2 No. of slits 8	T2
C 0.64~0.76	S3 No. of slits 16	T3
D 0.76~0.89	SS Optional	T4
E 0.89~1.02		T3A
F 1.02~1.15		T3B
G 1.15~1.28		T4A
H 1.28~1.41		T4B
I 1.41~1.54		
J 1.54~1.67		
K 1.67~1.80		

(mm)

55.56 × 0.1 × 19.05 × 45°

O.D.	Thickness ^{*4}	LD	Angle
55.56	0.060~0.200	19.05	θ
76.2		31.75	

(mm) (mm) (mm)

^{*4} Since the applicable ranges differ depending on the grit (mesh) size, bond and concentration, please contact your DISCO representative for details.